Immutability & Persistent Collections

Season: 6 · FP Foundations · Category: Functional · Level: Intermediate

Intent

- Share structure safely with Zazr Vector / NonEmptyVector.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Lego sets you can rebuild without melting old bricks.

When to Use

- Pure functional style in Java 25 + Zazr.
- The design force named in Intent shows up more than once.

When Not to Use

- Throwing exceptions everywhere is 'fine' for a script.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.functional.collections;

import dev.zazr.collection.NonEmptyVector;
import dev.zazr.collection.Vector;

public final class ZazrCollectionsDemo {
    private ZazrCollectionsDemo() {}

    public static Vector<Integer> doubleAll(Vector<Integer> xs) {
        return xs.map(n -> n * 2);
    }

    public static int maxScore(NonEmptyVector<Integer> scores) {
        return scores.max(Integer::compare);
    }

    public static Vector<Integer> buildFast(int n) {
        var b = Vector.<Integer>newBuilder();
        for (int i = 0; i < n; i++) b.add(i);
        return b.result();
    }
}
```

Modern Java 25 / Functional

- Primary examples use Zazr; VAVR shown in comparison episode.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: functional/collections/
```

Pitfalls

- Zazr rejects null inside Option/collections.
- Keep interpreters at the edge.

Interview Q&A

- Q: Interview: what problem does Immutability & Persistent Collections solve? A: Share structure safely with Zazr Vector / NonEmptyVector.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Share structure safely with Zazr Vector / NonEmptyVector.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Lego sets you can rebuild without melting old bricks. [0:40] Intent — State the design force in one ...